

Data Science Bootcamp

Full-Time





Overview

General Assembly's Data Science Bootcamp (DSB) is a transformative course that prepares students to break into data careers.

Created to transform your career in just 12 weeks, this full-time on-campus bootcamp will equip you with the tools and techniques needed to go from novice to competent and develop a data career full of opportunities.

You'll get skilled in everything from conducting an exploratory data analysis to running supervised and unsupervised machine learning models, before progressing on to learning about different types of neural networks and understanding how large language models work.

Once you complete this bootcamp, you'll present potential employers with an impressive resume highlighting your capabilities with the most in-demand tools they are looking for. You'll also have sought-after soft skills that will make you stand out.

Throughout this expert-designed course, you'll:

- Collect, extract, query, clean, and aggregate data for analysis.
- Gather, store, and organize data using SQL, APIs, and manage code using Git and GitHub.
- Perform visual and statistical analysis on data using Python and its associated libraries and tools.
- Craft and share compelling narratives through data visualization.
- Build and implement appropriate machine learning models and algorithms to evaluate data science problems spanning finance, public policy, and more.



- Clearly and effectively communicate to technical and nontechnical audiences through creating reports, meeting with stakeholders, and presenting results.
- Apply question, modeling, and validation problem-solving processes to data sets from various industries to provide insight into real-world problems and solutions.
- Prepare for the world of work, compiling a professional-grade portfolio of projects.
- Learn the principles of data engineering and how to manage data, from creating databases and pipelines to data governance.
- Learn how large language models (LLMs) work, and how to apply pretrained models to new data.
- Get an introduction to working with images in machine learning models through computer vision.
- Understand what generative AI is and techniques for effective prompt engineering.

Prerequisites

The course assumes no prior knowledge of either Python programming or data science. However, we recommend that students arrive with a mathematical foundation and familiarity with Python and programming fundamentals. Some students have a technical background, such as a degree in mathematics or computer science or work experience in research or analysis. Other students engage in self-learning to build a foundation ahead of class. Suitable work experience would be one or more years as a data analyst before starting the course.

Our [Admissions team](#) can discuss your background and learning goals to advise if this bootcamp is a good fit for you.



What you'll learn

Unit 1 Python and Data Science Fundamentals

Learn how to approach data science challenges and get acquainted with essential data science tools and techniques by learning Python, SQL, Git and GitHub.

- Learn about the data science lifecycle.
- Sharpen your presentation skills for both technical and nontechnical audiences.
- Learn the base Python you will need for data science to navigate data sources and collections.
- Run SQL in BigQuery.
- Learn to track changes and iterations using Git version control from your terminal.
- Share your code with others via GitHub.

Unit 2 Exploratory Data Analysis

Learn what exploratory data analysis (EDA) is and how to do it, including collecting and connecting data programmatically via APIs to add nuance to your models.

- Perform exploratory data analysis using pandas to turn a messy dataset into one that can be analyzed or used in a machine learning model.
- Generate visual and statistical analyses, using Python and its associated libraries and tools such as Matplotlib and seaborn.
- Collect and connect external data using APIs, in order to add nuance to your models.

Unit 3 Machine Learning Fundamentals

Build machine learning models. Implement regression and classification supervised learning models, and learn how to optimize them.

- Learn how to shape a data science problem.
- Use the scikit-learn and statsmodels libraries to build linear regression and logistic regression supervised machine learning models.
- Articulate the bias-variance trade-off as part of evaluating a model.
- Apply optimization and regularization for fitting and tuning models.
- Understand the principles of working with time series data.
- Optimize your code by leveraging generative AI tools.



Unit 4 **Advanced Modeling**

Explore the differences between supervised and unsupervised learning via clustering and dimensionality reduction.

- Build and evaluate ensemble models using decision trees, random forests, bagging, and boosting.
- Define clustering and its advantages and disadvantages as compared to classification models.
- Use Principal Component Analysis (PCA) to carry out dimensionality reduction.
- Dive into the math and theory for how gradient descent helps to optimize loss functions for machine learning models.
- Build a basic recommender system.

Unit 5 **Introduction to Data Engineering and Machine Learning Operations (MLOps)**

Explore the infrastructure used to support effective data science. As well as an introduction to data modeling, data pipelines, and data storage, learn how machine learning models are deployed within organizations. Practice sound data governance.

- Create a data model and a database.
 - Explain the forms and benefits of data normalization.
 - Differentiate between Extract, Transform, Load (ETL) and Extract, Load, Transform (ELT) processes when managing data pipelines.
 - Describe the tools used for data orchestration and the use cases for batch vs real-time processing.
 - Articulate the differences between data lakes and data warehouses, their architectures, use cases, and best practices for optimizing data storage.
 - Learn data governance principles that ensure security, integrity, and compliance.
 - Explain how machine learning models are deployed and managed in organizations using machine learning operations (MLOps).
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Unit 6 **Deep Learning and Large Language Models (LLMs)**

Dive into neural networks, comparing and contrasting different types of neural networks, and learn about transformers and large language models (LLMs): how they work and how to apply pretrained models to new data.

- Describe how neural networks work and common applications of deep learning.
- Use common deep learning toolkits: PyTorch, Hugging Face, and torchvision.
- Understand neural network architectures: feedforward networks, convolutional neural networks (CNNs), and recurrent neural networks (RNNs).
- Conduct transfer learning by fine-tuning pre-trained models for specific tasks (e.g., classification, object detection).
- Use a CNN to classify images (computer vision).
- Get acquainted with natural language processing (NLP)-the basis of large language models (LLMs)-to translate human language into a form that computers can use and apply to solve problems.
- Dive into transformer architecture and apply select models from the transformer family to solve a real-world problem.
- Develop generative AI skills through prompt engineering and chaining via LangChain.